MATHS MANTHAN
• TUITION CLASSES •
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Subject: Mathematics
Class: Class X (CBSE)

WEEKLY CHAPTER TEST – 03
Chapter 13: Surface Areas and Volumes

Time: 45 Minutes
MM: 20

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 objective questions (1 mark each).
2. Section B has 2 Short Answer questions (2 marks each). Section C has 2 questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (6 marks). Show all steps clearly.
4. Take $\pi = \frac{22}{7}$ unless specified otherwise. Show all steps and calculations clearly.

SECTION A: MULTIPLE CHOICE QUESTIONS (1 Mark Each) [4 Marks]

1. The number of solid spheres, each of diameter 6 cm, that can be made by melting a solid metal cylinder of height 45 cm and diameter 4 cm is: **[1]**

(a) (a) 3 (b) 5 (c) 4 (d) 6

2. The volumes of two spheres are in the ratio 64 : 27. The ratio of their surface areas is: **[1]**

(a) (a) 3 : 4 (b) 4 : 3 (c) 9 : 16 (d) 16 : 9

3. A right circular cylinder of radius r and height h ($h > 2r$) just encloses a sphere of diameter: **[1]**

(a) (a) r (b) $2r$ (c) h (d) $2h$

4. **Assertion (A):** If two solid right circular cylinders of the same height and base radii 3 cm and 4 cm are melted and recast into a cylinder of the same height, the radius of the new cylinder is 5 cm.
Reason (R): When two solid cylinders of equal height h and radii r_1, r_2 are recast into a cylinder of height h, the new base radius is $R = \sqrt{r_1^2 + r_2^2}$. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

5. A solid toy is in the form of a hemisphere surmounted by a right circular cone of the same base radius as that of the hemisphere. If the radius of the base of the cone is 21 cm and its volume is $\frac{2}{3}$ of the volume of the hemisphere, calculate the height of the cone. **[2]**

6. The largest possible right circular cone is carved out from a solid wooden cube of side 21 cm. Find the volume of the remaining solid. **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

7. A solid is composed of a cylinder with hemispherical ends. If the total length of the solid is 104 cm and the radius of each of the hemispherical ends is 7 cm, find the cost of polishing its surface at the rate of 10 per dm^2. **[3]**

8. From a solid cube of side 7 cm, a conical cavity of height 7 cm and radius 3 cm is hollowed out. Find the volume of the remaining solid. **[3]**

SECTION D: LONG ANSWER / HOTS QUESTION (6 Marks) [6 Marks]

9. A solid toy is in the form of a hemisphere surmounted by a right circular cone. The height of the cone is 4 cm and the diameter of the base is 8 cm.

(i) Determine the volume of the toy. (Use $\pi = 3.14$)
(ii) If a cube circumscribes the toy, find the difference of the volumes of the cube and the toy.
(iii) Find the total surface area of the toy.

[6]

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— **END OF QUESTION PAPER** —

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